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Exp-10 : Write a Python Program to Implement Expectation & Maximization Algorithm

Aim

To write a Python program to implement the Expectation-Maximization (EM) algorithm for clustering data into different groups.

Algorithm

1. Import the required Python libraries.
2. Create or load the input dataset.
3. Initialize the number of clusters.
4. Apply the Expectation step to calculate the probability of each data point belonging to each cluster.
5. Apply the Maximization step to update the cluster parameters.
6. Repeat the E-step and M-step until the model converges.
7. Display the final cluster assignments.

Program code

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.mixture import
GaussianMixture # Load the
Iris dataset
iris = load_iris()

X = iris.data

# Create the Gaussian Mixture Model (EM Algorithm)

gmm =
GaussianMixture(n_components=3,
random_state=42) # Train the model
```

gmm.fit(X)

Predict cluster
labels labels =
gmm.predict(X)
# Display
results
print("Cluster
Labels:")
print(labels)
print("\nMeans of Clusters:")
print(gmm.means_)

Sample Input
Data = [1.0, 1.2, 1.4, 5.0, 5.2, 5.4]
Number of clusters = 2

Output:

```
Cluster Labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 0 2 0 2 0 2
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 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0]

Means of Clusters:
[[6.54639415 2.94946365 5.48364578 1.98726565]
 [5.006      3.428      1.462      0.246     ]
 [5.9170732  2.77804839 4.20540364 1.29848217]]
```

Conclusion: The EM algorithm successfully grouped the given data into **two clusters** based on their probability distribution.